

Zero-Dimensional Energy Balance Model (EBM)

Tutorial 4

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1 Introduction

The Zero-Dimensional Energy Balance Model (EBM) is a simple model used to estimate Earth's equilibrium temperature and project future global mean surface temperature changes under different scenarios. This model works by balancing the incoming solar radiation with the outgoing longwave radiation from Earth's surface. The model is extended to include time-dependent changes, allowing for the simulation of future climate scenarios.

In this report, we will explore the following topics:

- The effect of changing the effective heat capacity on the future temperature projections.
- The effect of varying surface albedo on the future temperature trajectory.
- The effect of changing the initial temperature on the future temperature projection.

2 Methodology

The energy balance model was implemented using the following core equations:

$$S = \alpha F_{\text{in}} - \epsilon \sigma T^4$$

Where: - S is the change in temperature, - α is the surface albedo, - F_{in} is the incoming solar radiation, - ϵ is the emissivity, - σ is the Stefan-Boltzmann constant, and - T is the temperature.

The model was extended to time-dependent versions by introducing numerical methods, specifically forward difference methods, to simulate future temperature evolution.

3 Results

3.1 Question 1: Effect of Different Heat Capacities on Future Temperature Trajectories

In this section, we explore how the future temperature projections are affected by changing the effective heat capacity. Different values for the heat capacity components were considered. The resulting temperature trajectories for various cases are shown in the following plot.

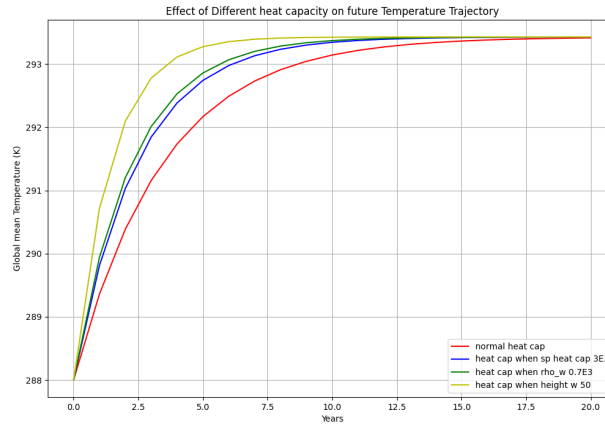


Figure 1: Effect of Different Heat Capacity on Future Temperature Trajectories

Observation: The plot shows how the temperature trajectories vary for different values of the heat capacity. As the heat capacity increases, the temperature change slows down, reflecting a higher resistance to temperature change.

3.2 Question 2: Effect of Varying Surface Albedo on Future Temperature Trajectories

Next, we examine how changing the surface albedo affects the future temperature projections. By using three different albedo values, we simulate the temperature trajectories.

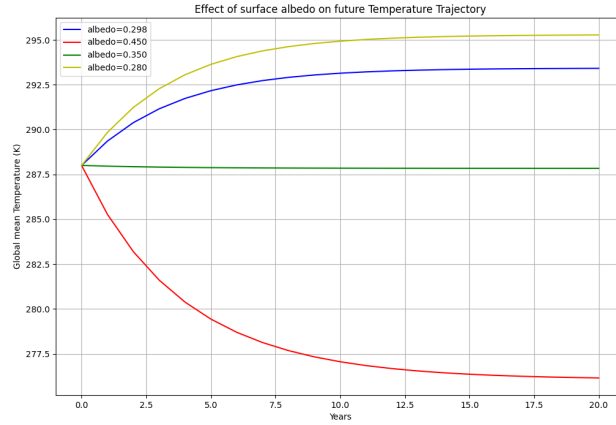


Figure 2: Effect of Surface Albedo on Future Temperature Trajectories

Observation: The temperature trajectory is significantly influenced by the value of surface albedo. A higher albedo reflects more sunlight, leading to a cooler temperature projection. Conversely, a lower albedo absorbs more sunlight, resulting in a warmer future temperature.

3.3 Question 3: Effect of Initial Temperature on Future Temperature Trajectories

Finally, we investigate how the initial temperature influences the future temperature projections, while keeping all other parameters constant. The following plot shows the future temperature trajectory for different initial temperatures.

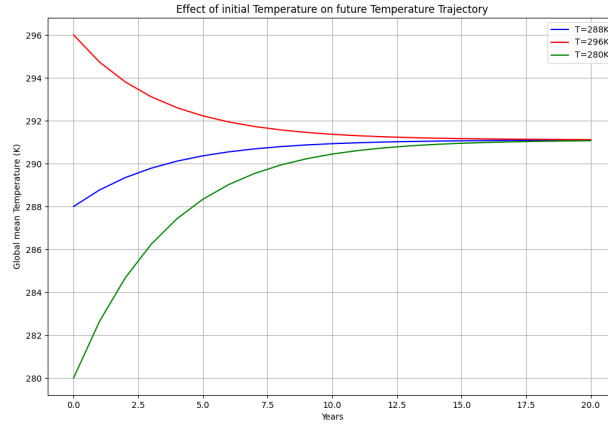


Figure 3: Effect of Initial Temperature on Future Temperature Trajectories

Observation: The initial temperature has a significant effect on the future temperature trajectory. Higher initial temperatures tend to result in a more rapid increase in global mean temperature over time, while lower initial temperatures lead to a slower increase.

4 Conclusion

Through this exercise, we have demonstrated the impact of three critical parameters (effective heat capacity, surface albedo, and initial temperature) on future temperature projections using the Zero-Dimensional Energy Balance Model. Each parameter plays a crucial role in determining how the Earth's temperature responds to different climatic forcings.